

Sensor Fusion and Data Association Problems

ELG 5218 – Uncertainty Evaluation in Engineering Measurements and Machine Learning

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PART A: CONCEPTUAL QUESTIONS (Simple to Intermediate)

A1. In precision-weighted fusion, why is it natural to work with precisions $\lambda_i = 1/\sigma_i^2$ rather than variances σ_i^2 ? What happens when one sensor has $\sigma_i \rightarrow \infty$?

Answer: Precisions are additive under independent fusion:

$$\lambda_f = \lambda_1 + \lambda_2 + \cdots + \lambda_N.$$

This is the information filter form: each sensor *adds* information. In contrast, variances combine via a non-linear harmonic-mean-like formula.

When $\sigma_i \rightarrow \infty$: $\lambda_i = 1/\sigma_i^2 \rightarrow 0$. The sensor contributes zero precision to the sum, so the fused estimate ignores it entirely. This is *automatic soft exclusion*—no thresholds or if-statements needed. A degraded sensor gets near-zero weight automatically.

A2. What is *data association*, and why is it needed in multi-target tracking?

Answer: Data association is the process of determining which measurement originated from which target (or from clutter). In single-sensor, single-target fusion (Block 1 of the slides), we assumed we *know* that y_1 and y_2 both came from target z . In multi-target tracking, the sensor returns a set of measurements $\{y^{(1)}, \dots, y^{(m)}\}$ and:

- Some may come from targets, others from clutter (false alarms).
- Each target is detected with probability $P_D < 1$ (missed detections possible).
- The origin of each measurement is *unknown*.

Example: Two aircraft fly close together, and the radar returns three blips. Without data association, we might assign the same blip to both tracks (double-counting) or assign a clutter blip to a track (incorporating noise as signal). Both lead to incorrect state estimates and overconfident covariances.

A3. *Classical gating:*

Answer:

Classical Gating	
Advantages	(1) No training data needed; threshold from χ^2 table. (2) Fully interpretable; gate = ellipsoidal contour at known confidence level.
Limitations	(1) Fixed threshold; cannot adapt to changing conditions. (2) Binary decision; loses information at the boundary.

A4. Define *aleatoric* and *epistemic* uncertainty in multi-sensor fusion.

Answer:

Aleatoric uncertainty is irreducible noise inherent in observations (sensor noise, environmental effects). Example: a radar in heavy rain produces noisy range measurements regardless of how much training data we have. This increases measurement noise $\mathbf{R}_t$, widening the posterior covariance.

Epistemic uncertainty is reducible uncertainty due to limited data or model capacity. Example: a CNN-based detector encounters a vehicle type never seen during training. More data of that type would reduce this uncertainty. Epistemic uncertainty affects the reliability of the model outputs (predictions may be confidently wrong).

Impact on fusion:

- Aleatoric uncertainty is captured by $\sigma_m^2(\mathbf{x})$ (heteroscedastic variance) and propagated through precision weighting. The Kalman filter naturally handles it via $\mathbf{R}$.
- Epistemic uncertainty requires ensemble disagreement, MC dropout variance, or evidential methods. In fusion, high epistemic uncertainty should trigger fallback to a classical method.

PART B: ANALYTICAL PROBLEMS — TRADITIONAL FUSION

B1. Two-Sensor Precision-Weighted Fusion

Answer:

(a) Precisions:

$$\lambda_1 = \frac{1}{\sigma_1^2} = \frac{1}{4} = 0.25, \quad \lambda_2 = \frac{1}{\sigma_2^2} = \frac{1}{1} = 1.0, \quad \boxed{\lambda_f = 0.25 + 1.0 = 1.25.}$$

(b) Weights:

$$w_1 = \frac{\lambda_1}{\lambda_f} = \frac{0.25}{1.25} = 0.20, \quad w_2 = \frac{\lambda_2}{\lambda_f} = \frac{1.0}{1.25} = 0.80, \quad \boxed{w_1 + w_2 = 1.0.} \checkmark$$

(c) Fused estimate and variance:

$$\hat{z} = 0.20 \times 4.8 + 0.80 \times 5.3 = 0.96 + 4.24 = \boxed{5.20.}$$

$$\sigma_f^2 = \frac{1}{\lambda_f} = \frac{1}{1.25} = \boxed{0.80}, \quad \sigma_f = 0.894.$$

(d) Check: $\sigma_f = 0.894 < \sigma_2 = 1.0 < \sigma_1 = 2.0$. $\checkmark$

The fused variance is always smaller than either individual sensor variance.

(e) Errors:

$$|y_1 - z| = |4.8 - 5| = 0.2, \quad |y_2 - z| = |5.3 - 5| = 0.3, \quad |\hat{z} - z| = |5.2 - 5| = \boxed{0.2.}$$

In this instance the fused error ties with sensor 1. On average (over many trials), the fused RMSE $\approx \sigma_f = 0.894$ beats $\sigma_2 = 1.0$.

(f) Information gain:

$$\Delta I = \frac{1}{2} \ln \frac{\sigma_1^2}{\sigma_f^2} = \frac{1}{2} \ln \frac{4}{0.8} = \frac{1}{2} \ln 5 = \frac{1}{2} (1.609) = \boxed{0.805 \text{ nats.}}$$

Interpretation: Adding sensor 2 provides 0.805 nats of information, reducing entropy from $H_1 = \frac{1}{2} \ln(2\pi e \cdot 4) = 1.919$ to $H_f = \frac{1}{2} \ln(2\pi e \cdot 0.8) = 1.115$ nats. Fusion always reduces entropy (or leaves it unchanged if the added sensor has infinite variance).

B2. Kalman Filter Multi-Sensor Update

Answer:

(a) Innovation:

$$\tilde{y} = y - \mathbf{H}\boldsymbol{\mu}_{t|t-1} = 12 - [1 \ 0] \begin{pmatrix} 10 \\ 2 \end{pmatrix} = 12 - 10 = \boxed{2}.$$

(b) Innovation covariance:

$$S = \mathbf{H}\boldsymbol{\Sigma}_{t|t-1}\mathbf{H}^T + R = [1 \ 0] \begin{pmatrix} 5 & 1 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} + 4 = 5 + 4 = \boxed{9}.$$

(c) Kalman gain:

$$\mathbf{K} = \boldsymbol{\Sigma}_{t|t-1}\mathbf{H}^T S^{-1} = \begin{pmatrix} 5 & 1 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} \cdot \frac{1}{9} = \begin{pmatrix} 5 \\ 1 \end{pmatrix} \cdot \frac{1}{9} = \boxed{\begin{pmatrix} 0.556 \\ 0.111 \end{pmatrix}}.$$

(d) Updated mean:

$$\boldsymbol{\mu}_{t|t} = \begin{pmatrix} 10 \\ 2 \end{pmatrix} + \begin{pmatrix} 0.556 \\ 0.111 \end{pmatrix} \cdot 2 = \boxed{\begin{pmatrix} 11.11 \\ 2.22 \end{pmatrix}}.$$

Interpretation: Position moved toward GPS reading (from 10 to 11.11). Velocity also adjusted upward (from 2.0 to 2.22) because the cross-covariance $P_{12} = 1$ links position and velocity.

(e) Updated covariance:

$$\begin{aligned} \boldsymbol{\Sigma}_{t|t} &= (\mathbf{I} - \mathbf{K}\mathbf{H})\boldsymbol{\Sigma}_{t|t-1} = \left(\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} - \begin{pmatrix} 0.556 \\ 0.111 \end{pmatrix} [1 \ 0] \right) \begin{pmatrix} 5 & 1 \\ 1 & 2 \end{pmatrix} \\ &= \begin{pmatrix} 0.444 & 0 \\ -0.111 & 1 \end{pmatrix} \begin{pmatrix} 5 & 1 \\ 1 & 2 \end{pmatrix} = \boxed{\begin{pmatrix} 2.22 & 0.44 \\ 0.44 & 1.89 \end{pmatrix}}. \end{aligned}$$

(f) Determinant check:

$$\det(\boldsymbol{\Sigma}_{t|t-1}) = 5 \cdot 2 - 1 \cdot 1 = 9, \quad \det(\boldsymbol{\Sigma}_{t|t}) = 2.22 \cdot 1.89 - 0.44^2 = 4.20 - 0.19 = 4.01.$$

Indeed $4.01 < 9$. ✓ *Meaning:* The GPS measurement reduced the state uncertainty ellipse area by a factor of ~ 2.2 .

(g) Sequential IMU update. Starting from $(\boldsymbol{\mu}_{t|t}, \boldsymbol{\Sigma}_{t|t})$ above, $\mathbf{H}_2 = [0 \ 1]$, $R_2 = 1$, $y_2 = 2.5$:

$$\text{Innovation: } \tilde{y}_2 = 2.5 - [0 \ 1]\boldsymbol{\mu}_{t|t} = 2.5 - 2.22 = 0.28.$$

$$S_2 = [0 \ 1]\boldsymbol{\Sigma}_{t|t}[0 \ 1]^T + 1 = 1.89 + 1 = 2.89.$$

$$\mathbf{K}_2 = \boldsymbol{\Sigma}_{t|t}[0 \ 1]^T / 2.89 = \begin{pmatrix} 0.44 \\ 1.89 \end{pmatrix} / 2.89 = \begin{pmatrix} 0.152 \\ 0.654 \end{pmatrix}.$$

$$\boldsymbol{\mu}_{\text{final}} = \begin{pmatrix} 11.11 \\ 2.22 \end{pmatrix} + \begin{pmatrix} 0.152 \\ 0.654 \end{pmatrix} \cdot 0.28 = \boxed{\begin{pmatrix} 11.15 \\ 2.40 \end{pmatrix}}.$$

$$\boldsymbol{\Sigma}_{\text{final}} = (\mathbf{I} - \mathbf{K}_2\mathbf{H}_2)\boldsymbol{\Sigma}_{t|t} = \begin{pmatrix} 2.15 & 0.16 \\ 0.16 & 0.65 \end{pmatrix}.$$

Interpretation: After both GPS and IMU updates, uncertainty is substantially reduced. The velocity covariance dropped from 2.0 to 0.65.

B3. Elliptical Gating with Mahalanobis Distance

Learning objective: Apply elliptical gating using the Mahalanobis distance and interpret why correlated uncertainty changes the acceptance region.

Solution.

We are given the predicted measurement

$$\hat{\mathbf{y}} = \begin{pmatrix} 10 \\ 5 \end{pmatrix},$$

the innovation covariance

$$\mathbf{S} = \begin{pmatrix} 4.25 & 3.0 \\ 3.0 & 5.25 \end{pmatrix}, \quad \mathbf{S}^{-1} = \begin{pmatrix} 0.394 & -0.225 \\ -0.225 & 0.319 \end{pmatrix},$$

and the gating threshold

$$\gamma = \chi_2^2(0.99) = 9.21.$$

The candidate measurements are

$$P_1 : (13, 5), \quad P_2 : (10, 8), \quad P_3 : (12, 7), \quad P_4 : (15, 1).$$

A point is inside the gate if

$$d_M^2 = \tilde{\mathbf{y}}^T \mathbf{S}^{-1} \tilde{\mathbf{y}} \leq \gamma.$$

(a) Compute the innovation vectors

The innovation is

$$\tilde{\mathbf{y}} = \mathbf{y} - \hat{\mathbf{y}}.$$

Therefore,

$$\tilde{\mathbf{y}}_1 = \begin{pmatrix} 13 \\ 5 \end{pmatrix} - \begin{pmatrix} 10 \\ 5 \end{pmatrix} = \begin{pmatrix} 3 \\ 0 \end{pmatrix},$$

$$\tilde{\mathbf{y}}_2 = \begin{pmatrix} 10 \\ 8 \end{pmatrix} - \begin{pmatrix} 10 \\ 5 \end{pmatrix} = \begin{pmatrix} 0 \\ 3 \end{pmatrix},$$

$$\tilde{\mathbf{y}}_3 = \begin{pmatrix} 12 \\ 7 \end{pmatrix} - \begin{pmatrix} 10 \\ 5 \end{pmatrix} = \begin{pmatrix} 2 \\ 2 \end{pmatrix},$$

$$\tilde{\mathbf{y}}_4 = \begin{pmatrix} 15 \\ 1 \end{pmatrix} - \begin{pmatrix} 10 \\ 5 \end{pmatrix} = \begin{pmatrix} 5 \\ -4 \end{pmatrix}.$$

(b) Compute the squared Mahalanobis distances

Point P_1 :

$$d_{M,1}^2 = \begin{pmatrix} 3 & 0 \end{pmatrix} \begin{pmatrix} 0.394 & -0.225 \\ -0.225 & 0.319 \end{pmatrix} \begin{pmatrix} 3 \\ 0 \end{pmatrix}.$$

Since only the first component is nonzero,

$$d_{M,1}^2 = 3^2(0.394) = 9(0.394) = 3.546.$$

Point P_2 :

$$d_{M,2}^2 = \begin{pmatrix} 0 & 3 \end{pmatrix} \begin{pmatrix} 0.394 & -0.225 \\ -0.225 & 0.319 \end{pmatrix} \begin{pmatrix} 0 \\ 3 \end{pmatrix}.$$

Since only the second component is nonzero,

$$d_{M,2}^2 = 3^2(0.319) = 9(0.319) = 2.871.$$

Point P_3 :

$$d_{M,3}^2 = \begin{pmatrix} 2 & 2 \end{pmatrix} \begin{pmatrix} 0.394 & -0.225 \\ -0.225 & 0.319 \end{pmatrix} \begin{pmatrix} 2 \\ 2 \end{pmatrix}.$$

First compute

$$\mathbf{S}^{-1}\tilde{\mathbf{y}}_3 = \begin{pmatrix} 0.394(2) - 0.225(2) \\ -0.225(2) + 0.319(2) \end{pmatrix} = \begin{pmatrix} 0.338 \\ 0.188 \end{pmatrix}.$$

Then

$$d_{M,3}^2 = \begin{pmatrix} 2 & 2 \end{pmatrix} \begin{pmatrix} 0.338 \\ 0.188 \end{pmatrix} = 2(0.338) + 2(0.188) = 1.052.$$

Point P_4 :

$$d_{M,4}^2 = \begin{pmatrix} 5 & -4 \end{pmatrix} \begin{pmatrix} 0.394 & -0.225 \\ -0.225 & 0.319 \end{pmatrix} \begin{pmatrix} 5 \\ -4 \end{pmatrix}.$$

First compute

$$\mathbf{S}^{-1}\tilde{\mathbf{y}}_4 = \begin{pmatrix} 0.394(5) - 0.225(-4) \\ -0.225(5) + 0.319(-4) \end{pmatrix} = \begin{pmatrix} 1.97 + 0.90 \\ -1.125 - 1.276 \end{pmatrix} = \begin{pmatrix} 2.87 \\ -2.401 \end{pmatrix}.$$

Then

$$d_{M,4}^2 = \begin{pmatrix} 5 & -4 \end{pmatrix} \begin{pmatrix} 2.87 \\ -2.401 \end{pmatrix} = 5(2.87) + (-4)(-2.401) = 14.35 + 9.604 = 23.954.$$

Hence,

$d_{M,1}^2 = 3.546, \quad d_{M,2}^2 = 2.871, \quad d_{M,3}^2 = 1.052, \quad d_{M,4}^2 = 23.954.$
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(c) **Decide whether each point is inside the gate**

The rule is

$$d_M^2 \leq 9.21.$$

Therefore,

$$P_1 : 3.546 < 9.21 \Rightarrow \text{inside},$$

$$P_2 : 2.871 < 9.21 \Rightarrow \text{inside},$$

$$P_3 : 1.052 < 9.21 \Rightarrow \text{inside},$$

$$P_4 : 23.954 > 9.21 \Rightarrow \text{outside}.$$

So,

$$\boxed{P_1, P_2, P_3 \text{ are inside the gate, and } P_4 \text{ is outside.}}$$

(d) **Why Mahalanobis distance differs from Euclidean distance**

The Euclidean distance only measures geometric distance:

$$\|\tilde{\mathbf{y}}\| = \sqrt{\tilde{\mathbf{y}}_1^2 + \tilde{\mathbf{y}}_2^2}.$$

It ignores the covariance structure of the uncertainty.

The Mahalanobis distance instead uses

$$d_M^2 = \tilde{\mathbf{y}}^T \mathbf{S}^{-1} \tilde{\mathbf{y}},$$

so it accounts for:

- different uncertainty scales in different directions, and
- correlation between measurement components.

Thus, the acceptance region is not a circle but an ellipse shaped by $\mathbf{S}$. A point may look moderately far in Euclidean distance, but if it lies in a direction of high uncertainty it can still be acceptable. Conversely, a point in a direction of low uncertainty can be penalized more strongly.

In this example, P_4 is not only far in Euclidean distance, but it also lies in a direction that is strongly penalized by the covariance geometry, so its Mahalanobis distance becomes very large.

(e) **Lowering the gate to $P_G = 0.95$**

For $P_G = 0.95$, the threshold becomes

$$\gamma = \chi_2^2(0.95) = 5.99.$$

Now compare each value to 5.99:

$$d_{M,1}^2 = 3.546 < 5.99, \quad d_{M,2}^2 = 2.871 < 5.99, \quad d_{M,3}^2 = 1.052 < 5.99, \quad d_{M,4}^2 = 23.954 > 5.99.$$

So the result is unchanged:

$$\boxed{P_1, P_2, P_3 \text{ remain inside, and } P_4 \text{ remains outside.}}$$

The trade-off is:

- a **larger gate** accepts more true measurements, but also more clutter and false associations;

- a **smaller gate** rejects more clutter, but increases the risk of discarding a correct measurement.

B4. JPDA: Joint Association Probabilities

Answer:

(a) Unnormalised scores:

$$w(\Theta_1) = P_D^2 \cdot \mathcal{L}_{11} \cdot \mathcal{L}_{22} = 0.9^2 \times 0.50 \times 0.40 = 0.81 \times 0.20 = \boxed{0.162.}$$

$$w(\Theta_2) = P_D^2 \cdot \mathcal{L}_{12} \cdot \mathcal{L}_{21} = 0.81 \times 0.05 \times 0.08 = 0.81 \times 0.004 = \boxed{0.00324.}$$

(b) Normalised:

$$P(\Theta_1) = \frac{0.162}{0.162 + 0.00324} = \frac{0.162}{0.16524} = \boxed{0.980.}$$

$$P(\Theta_2) = \frac{0.00324}{0.16524} = \boxed{0.020.}$$

(c) Marginal $\beta_{1,1}$ (Track 1 gets $\mathbf{y}^{(1)}$): This occurs in Θ_1 only:

$$\boxed{\beta_{1,1} = P(\Theta_1) = 0.980.}$$

Track 1 is 98% confident that $\mathbf{y}^{(1)}$ is its measurement.

(d) Marginal $\beta_{2,1}$ (Track 2 gets $\mathbf{y}^{(1)}$): This occurs in Θ_2 only:

$$\boxed{\beta_{2,1} = P(\Theta_2) = 0.020.}$$

(e) Check: $\beta_{1,1} + \beta_{2,1} = 0.980 + 0.020 = 1.00 \leq 1. \checkmark$

In PDAF (no inter-target constraint), Track 1 and Track 2 compute their β values independently. Both could have $\beta_{i,1} \approx 0.9$ for the same measurement, violating the one-to-one constraint. In JPDA, the joint probability structure forces measurement $\mathbf{y}^{(1)}$ to be “claimed” mostly by Track 1, leaving only 2% for Track 2. This prevents both tracks from pulling toward the same measurement (track coalescence).

(f) If $\mathcal{L}_{11} = \mathcal{L}_{21} = 0.30$ (ambiguous):

$$w(\Theta_1) = 0.81 \times 0.30 \times 0.40 = 0.0972,$$

$$w(\Theta_2) = 0.81 \times 0.05 \times 0.30 = 0.01215.$$

$$P(\Theta_1) = \frac{0.0972}{0.1094} = 0.889, \quad P(\Theta_2) = 0.111.$$

Now $\beta_{1,1} = 0.889$ and $\beta_{2,1} = 0.111$: Track 2’s claim on $\mathbf{y}^{(1)}$ increased from 2% to 11%. When the measurement fits both tracks similarly, JPDA spreads the probability more evenly, increasing the spread term and reflecting genuine association uncertainty.

PART C: ANALYTICAL PROBLEMS — DEEP FUSION

C1. Cosine Similarity vs. Euclidean Distance for Embeddings

Answer:

(a) Euclidean distances:

$$\|\mathbf{a} - \mathbf{b}\| = \sqrt{(4 - 1.2)^2 + (3 - 0.9)^2} = \sqrt{7.84 + 4.41} = \sqrt{12.25} = \boxed{3.50}.$$

$$\|\mathbf{a} - \mathbf{c}\| = \sqrt{(4 + 2)^2 + (3 - 3.5)^2} = \sqrt{36 + 0.25} = \boxed{6.02}.$$

By Euclidean distance, $\mathbf{a}$ is closer to $\mathbf{b}$ (3.50) than to $\mathbf{c}$ (6.02). *However*, the dark car $\mathbf{b}$ (3.50) is farther from the bright car than $\mathbf{c}$ (truck) is from $\mathbf{b}$: $\|\mathbf{b} - \mathbf{c}\| = \sqrt{(3.2)^2 + (2.6)^2} = 4.12$. So Euclidean does give the right ranking here, but the margin is misleading since $\mathbf{a}$ and $\mathbf{b}$ are the *same object*.

(b) Cosine similarities:

$$\text{Sim}(\mathbf{a}, \mathbf{b}) = \frac{4(1.2) + 3(0.9)}{\sqrt{16 + 9}\sqrt{1.44 + 0.81}} = \frac{4.8 + 2.7}{5 \times 1.5} = \frac{7.5}{7.5} = \boxed{1.000}.$$

$$\text{Sim}(\mathbf{a}, \mathbf{c}) = \frac{4(-2) + 3(3.5)}{5 \times \sqrt{4 + 12.25}} = \frac{-8 + 10.5}{5 \times 4.03} = \frac{2.5}{20.16} = \boxed{0.124}.$$

- (c) Cosine similarity correctly identifies $\mathbf{a}$ and $\mathbf{b}$ as identical objects (Sim = 1.0, perfect match) because they point in the same direction. The magnitude difference (bright vs. dark) is factored out by normalisation. Euclidean distance penalises the magnitude change.
- (d) In high-dimensional spaces ($d = 256$), Euclidean distances between random points concentrate around $\sqrt{d}$ —all pairs become nearly equidistant (“curse of dimensionality”). Cosine similarity measures the *angle* between vectors, which does not suffer from this concentration, because normalisation removes the magnitude component that dominates in high dimensions.

C2. Heteroscedastic PoE Fusion under Sensor Corruption

Answer:

(a) Normal conditions:

$$\sigma_f^{-2} = \frac{1}{0.5} + \frac{1}{0.8} = 2.0 + 1.25 = 3.25, \quad \sigma_f^2 = \frac{1}{3.25} = \boxed{0.308}.$$

$$\mu_f = 0.308 \left(\frac{4.8}{0.5} + \frac{5.1}{0.8} \right) = 0.308(9.6 + 6.375) = 0.308 \times 15.975 = \boxed{4.92}.$$

(b) Weights (fraction of total precision):

$$w_A = \frac{2.0}{3.25} = \boxed{0.615}, \quad w_B = \frac{1.25}{3.25} = \boxed{0.385}.$$

IMU contributes $\sim 62\%$, camera $\sim 38\%$.

(c) Corrupted conditions:

$$\sigma_f^{-2} = \frac{1}{0.5} + \frac{1}{12.0} = 2.0 + 0.0833 = 2.0833, \quad \sigma_f^2 = \frac{1}{2.0833} = \boxed{0.480}.$$

$$\mu_f = 0.480 \left(\frac{4.8}{0.5} + \frac{7.2}{12.0} \right) = 0.480(9.6 + 0.6) = 0.480 \times 10.2 = \boxed{4.90}.$$

(d) Corrupted weights:

$$w_A = \frac{2.0}{2.0833} = \boxed{0.960}, \quad w_B = \frac{0.0833}{2.0833} = \boxed{0.040}.$$

Camera now contributes only 4% of the information.

(e) The fused estimate shifted from 4.92 to 4.90—almost no change despite the camera reading being 7.2 (wildly wrong). The shift is small because the camera’s large predicted variance ($\sigma^2 = 12$) gives it near-zero precision, so it contributes almost no weight. The precision-weighting mechanism automatically soft-excludes the unreliable sensor.

(f) Homoscedastic case ($\sigma_A^2 = \sigma_B^2 = 1$ always):

$$\mu_f = \frac{\mu_A + \mu_B}{2} = \frac{4.8 + 7.2}{2} = \boxed{6.0}.$$

This is 1.2 away from the true value—the corrupted camera drags the estimate significantly. Without learned variance, the fusion system has no mechanism to detect degradation. This is why **heteroscedastic uncertainty is essential for robust fusion**.

PART D: INTERPRETATION AND PRACTICAL QUESTIONS

D1. Interpreting Mixture-of-Experts Gate Weights

Answer:

- (a) As modality B becomes more corrupted, the gating network recognises (from its input features) that B’s signal is unreliable and shifts weight to A. This is the learned analogue of precision weighting: corrupted sensor $\Rightarrow$ low reliability $\Rightarrow$ low gate weight.
- (b) At $\sigma = 20$, RMSE is $0.65 > 0.58$ (A-only). The 5% residual gate weight on B still lets some corrupted signal through. Also, the gating network itself may be slightly confused by the extreme corruption (it was trained with dropout noise up to some level, but $\sigma = 20$ may be beyond training distribution). The gating function is imperfect.
- (c) Without modality dropout training, the network would have learned to always rely on both modalities. Under corruption, it would not know how to shift weight to A—the gate weights would stay near (0.58, 0.42) regardless. RMSE under corruption would be *much worse*, similar to early fusion: ~ 5 – 10 at high corruption. The network fails silently.
- (d) In classical precision-weighted fusion, $w_m \propto 1/\sigma_m^2$ —noisier sensors get lower weight. The MoE gate $g_m(\mathbf{x})$ learns a similar function from data. If the gating network is well-calibrated, it approximates $g_m \approx \lambda_m / \sum_k \lambda_k$, where λ_m is the precision of modality m . The difference is that classical weights require known σ_m^2 , while the MoE learns them from data.

D2. MC Dropout and Epistemic Uncertainty for Fusion

Answer:

- (a) MC dropout variance primarily measures **epistemic uncertainty**. By running multiple stochastic forward passes with different dropout masks, the variance captures how sensitive the model’s predictions are to its internal representation. When the model is uncertain about its parameters (epistemic), different dropout masks produce different outputs.
- (b) OOD variance is much larger (2.8 vs. 0.12) because the model has not seen similar inputs during training. Different parts of the network disagree about how to process unfamiliar inputs, leading to high variance across dropout samples. The model essentially “doesn’t know what it doesn’t know” but the disagreement reveals this.
- (c) Deep ensembles train M networks with different random initialisations. Each converges to a different local minimum. For in-distribution inputs, all minima agree (low variance). For OOD inputs, the minima disagree more strongly than dropout masks do (which share the same trained weights). Hence ensembles give 4.1 vs. MC dropout’s 2.8—a stronger OOD signal.
- (d) Use a threshold on the predictive variance. If $\text{variance} > \text{threshold}$ (calibrated on a validation set), flag the input as unreliable and fall back to the classical Kalman filter (which uses physics-based models and does not suffer from OOD in the same way). This creates a **two-tier system**: deep fusion when confident, classical when uncertain.

However, if we want to dig deeper, then it is crucial to distinguish between two different types of out-of-distribution (OOD) behavior in this context:

- (a) **Deep Learning OOD (violated training assumptions)**. A neural network becomes uncertain when the data no longer resembles its training set. For example, a camera-based fusion system might encounter heavy snow, an unusually shaped vehicle, or lens glare. In this case, the *appearance* of the data is OOD for the neural network, but the *physical kinematics* of the vehicle (e.g., velocity, mass, momentum) remain valid. Here, falling back to a classical Kalman filter (e.g., relying on radar, IMU, or GPS) can still work well because basic physical motion models (such as constant velocity or constant turn rate) continue to apply.
- (b) **Physical/System OOD (violated Kalman filter assumptions)**. Sometimes the anomalous event is not merely visual but physical. Examples include:
 - A hardware failure (a sensor becomes misaligned or fails entirely).
 - A fundamental change in operating conditions (e.g., the vehicle hits black ice, so the tire–friction model assumed by an Extended Kalman Filter becomes invalid).
 - Extreme evasive maneuvers that violate the assumed state transition model F or process-noise covariance Q .

In these cases, the OOD event breaks the assumptions of the physics-based model. If the Kalman filter blindly takes over, it can also fail because its internal model no longer matches reality.

D3. Missing Modality Robustness

Answer:

- (a) **Early fusion** (concatenation) will fail catastrophically. The network received zeros where it expects features, producing garbage activations throughout. It has no mechanism to detect or route around the missing input. Late fusion will degrade somewhat (modality B’s head produces a bad prediction), but the combination layer may partially compensate. **Gated MoE** handles it best—the gate shifts weight to A.
- (b) In the information filter, the update adds each modality’s precision contribution:

$$\mathbf{\Lambda}_{t|t} = \mathbf{\Lambda}_{t|t-1} + \sum_{m=1}^M \mathbf{H}_m^T \mathbf{R}_m^{-1} \mathbf{H}_m.$$

If modality m is missing, simply omit that term: $\mathbf{R}_m^{-1} \rightarrow 0$. The result is the same as if that sensor had infinite noise. No imputation, no special logic—the math handles it naturally.

- (c) Modality dropout during training repeatedly simulates the “missing sensor” scenario. The network learns to produce correct outputs even when some inputs are zero. The gating network learns to detect which modalities are absent/corrupted (from the zero-valued features) and shift weight accordingly. This is the learned analogue of omitting the precision term.

PART E: SYNTHESIS PROBLEMS

E1. Correlated Sensors and Covariance Intersection

Answer:

- (a) Independent case:

$$\sigma_{f,\text{naive}}^2 = \frac{\sigma_1^2 \sigma_2^2}{\sigma_1^2 + \sigma_2^2} = \frac{4 \times 4}{4 + 4} = \frac{16}{8} = \boxed{2.0}.$$

- (b) Correct fused variance with $\Sigma_{12} = 1$:

$$\sigma_f^2 = \frac{\sigma_1^2 \sigma_2^2 - \Sigma_{12}^2}{\sigma_1^2 + \sigma_2^2 - 2\Sigma_{12}} = \frac{16 - 1}{8 - 2} = \frac{15}{6} = \boxed{2.50}.$$

- (c) Underestimation:

$$\text{Ratio} = \frac{\sigma_{f,\text{naive}}^2}{\sigma_f^2} = \frac{2.0}{2.5} = 0.80.$$

The naive formula underestimates uncertainty by $\boxed{20\%}$. The filter believes it is more accurate than it actually is.

- (d) Covariance Intersection with $\omega = 0.5$:

$$\sigma_f^{-2} = 0.5 \times \frac{1}{4} + 0.5 \times \frac{1}{4} = 0.125 + 0.125 = 0.25, \quad \sigma_f^2 = \frac{1}{0.25} = \boxed{4.0}.$$

This is *larger* than the correct value of 2.5. CI is conservative—it overestimates uncertainty rather than underestimating it.

- (e) In safety-critical systems (air traffic control, autonomous driving), an overconfident filter may:

- Make the gate too small, missing true measurements.
- Report unrealistically tight confidence intervals, leading to dangerous decisions.
- Diverge: the predicted covariance shrinks while the actual error grows, until the innovation exceeds the gate permanently.

A conservative (larger) covariance is safer: the gate stays wide enough, confidence intervals are honest, and the filter remains stable. CI provides this guarantee for any unknown positive correlation.

E2. End-to-End Comparison: Stress Testing Fusion Methods

Answer:

- At $\sigma_{\text{corrupt}} = 0$, all multi-modal methods beat A-only because they combine information from *two* modalities. By precision additivity ($\lambda_f = \lambda_A + \lambda_B > \lambda_A$), the fused variance is smaller than either individual variance. This is the fundamental benefit of sensor fusion.
- Early fusion crosses the A-only baseline between $\sigma_{\text{corrupt}} = 1$ and $\sigma_{\text{corrupt}} = 2$: at $\sigma = 1$, $\text{RMSE} = 0.48 < 0.58$; at $\sigma = 2$, $\text{RMSE} = 0.72 > 0.58$. So early fusion becomes *harmful* at roughly $\sigma_{\text{corrupt}} \approx 1.5$. Beyond this point, having modality B is worse than not having it—because early fusion has no mechanism to downweight it.
- Gated MoE approaches but doesn't reach the A-only baseline because: (i) the gate never reaches exactly $g_B = 0$ (the softmax saturates but never reaches zero); (ii) the small residual weight on B lets some noise through; (iii) the gating network itself introduces a small amount of parameter uncertainty. Still, degradation is graceful: RMSE increases from 0.32 to 0.58, a $\sim 2\times$ increase, compared to early fusion's $\sim 27\times$ increase.
- Heteroscedastic PoE is best because it implements the same *precision-weighting* principle as classical fusion (B1): each modality's predicted variance σ_m^2 directly determines its weight $\propto 1/\sigma_m^2$. When the camera is corrupted, it learns to output large σ^2 , giving near-zero precision. This is mathematically identical to the Kalman filter information form where a missing/degraded sensor contributes $\mathbf{R}_m^{-1} \rightarrow 0$.
- For a safety-critical autonomous driving system, I would deploy the **heteroscedastic PoE**, because it gives the best overall accuracy in the clean setting and also degrades most gracefully under corruption. Its behavior is interpretable: unreliable modalities are explicitly assigned lower precision, which is desirable in safety-critical settings.

An additional safeguard should be added on top of the fusion model. A good choice is a **modality-health monitor** or **OOD/corruption detector** that checks whether a sensor is behaving abnormally. If a modality is flagged as unreliable, the system can:

- reduce or remove its contribution to fusion,
- trigger a fallback mode,
- raise an uncertainty-aware warning to downstream planning or control modules.

This creates a two-layer defense: robust fusion in normal operation, and explicit fault handling under severe sensor degradation.

Engineering takeaway. A good multimodal system should satisfy three properties:

1. it should outperform unimodal baselines when all sensors are healthy,
2. it should degrade gracefully when one modality is corrupted,
3. it should approach the best remaining unimodal expert when the corrupted modality becomes unusable.

Early fusion fails this test, while gated and uncertainty-aware fusion methods are much better suited for real deployments.

1. F1. Sequential Kalman fusion: IMU every step, GPS every 5 steps

Answer:

- (a) With IMU-only measurements, the filter observes v_k directly but not p_k . Since p_k evolves by integration,

$$p_k = p_{k-1} + \Delta t v_{k-1} + \text{process noise},$$

any small bias/noise in velocity (and process noise in dynamics) accumulates over time into position. In observability terms, position is only indirectly constrained through the motion model, so its uncertainty tends to grow (or at least not be strongly corrected) without absolute position fixes.

- (b) Each step, prediction increases uncertainty:

$$\mathbf{P}_{k|k-1} = \mathbf{F} \mathbf{P}_{k-1|k-1} \mathbf{F}^\top + \mathbf{Q},$$

so $\text{tr}(\mathbf{P})$ goes up. The IMU update reduces uncertainty mainly in v (and partially in p via cross-covariance P_{pv}). Every 5th step, the GPS update directly measures p and causes a sharper covariance reduction, hence a periodic drop (sawtooth). The reported small final trace $\text{tr}(\mathbf{P}_T) = 0.3681$ is consistent with frequent velocity corrections plus occasional absolute position corrections.

- (c) Let the two measurements (when GPS is present) be conditionally independent given $\mathbf{x}_k$. Define a stacked measurement

$$\mathbf{z}_k = \begin{bmatrix} z_k^{\text{imu}} \\ z_k^{\text{gps}} \end{bmatrix}, \quad \mathbf{H} = \begin{bmatrix} \mathbf{H}_{\text{imu}} \\ \mathbf{H}_{\text{gps}} \end{bmatrix}, \quad \mathbf{R} = \begin{bmatrix} R_{\text{imu}} & 0 \\ 0 & R_{\text{gps}} \end{bmatrix}.$$

Then the standard Kalman update with $(\mathbf{H}, \mathbf{R})$ yields the same posterior as two sequential updates because the joint likelihood factorizes:

$$p(\mathbf{z}_k | \mathbf{x}_k) = p(z_k^{\text{imu}} | \mathbf{x}_k) p(z_k^{\text{gps}} | \mathbf{x}_k).$$

In information form, independent measurements add information:

$$\mathbf{Y}^+ = \mathbf{Y}^- + \mathbf{H}_{\text{imu}}^\top R_{\text{imu}}^{-1} \mathbf{H}_{\text{imu}} + \mathbf{H}_{\text{gps}}^\top R_{\text{gps}}^{-1} \mathbf{H}_{\text{gps}},$$

which is exactly what sequential updating performs.

2. F2. Product of Experts fusion and missing modality

Answer:

For Gaussian experts, PoE is Gaussian with precision additivity:

$$\lambda_m = \frac{1}{\sigma_m^2}, \quad \lambda_f = \sum_m \lambda_m, \quad \sigma_f^2 = \frac{1}{\lambda_f}, \quad \mu_f = \sigma_f^2 \sum_m \lambda_m \mu_m.$$

(a)

$$\lambda_{\text{IMU}} = 2.0, \quad \lambda_{\text{GPS}} = \frac{1}{1.5} = 0.6667, \quad \lambda_{\text{Cam}} = \frac{1}{0.8} = 1.25.$$

So

$$\lambda_f = 2.0 + 0.6667 + 1.25 = 3.9167, \quad \sigma_f^2 = \frac{1}{3.9167} = 0.2553,$$

$$\mu_f = 0.2553 (2.0 \cdot 4.8 + 0.6667 \cdot 5.3 + 1.25 \cdot 5.1) = 4.9809.$$

(These match the notebook output: $\mu_f \approx 4.9809$, $\sigma_f^2 \approx 0.2553$.)

(b) Dropping GPS removes its precision contribution:

$$\lambda_f = 2.0 + 1.25 = 3.25, \quad \sigma_f^2 = \frac{1}{3.25} = 0.3077,$$

$$\mu_f = 0.3077 (2.0 \cdot 4.8 + 1.25 \cdot 5.1) = 4.9154.$$

Uncertainty increases because there is less total information (smaller λ_f), and the mean shifts because the GPS expert no longer pulls the fused estimate toward 5.3; the result moves toward the remaining experts (IMU and Camera).

3. F3. Crossing targets: why GNN, PDAF, JPDA have similar RMSE here

Answer:

(a) In this run, the scene is relatively clean: P_D is high and clutter is low, so at each step the *correct* measurement is typically the nearest one inside the gate. Therefore:

- GNN’s hard assignment almost always picks the correct measurement.
- PDAF’s association probabilities β_j become nearly one-hot (one $\beta_j \approx 1$), making it behave like a standard KF update.
- JPDA collapses to the dominant joint hypothesis, producing essentially the same update.

Hence similar RMSE.

(b) The table is warning about ambiguity:

- **GNN (hard assignment):** if two measurements are similarly plausible (near crossing, in clutter), a single wrong hard assignment can cause a *track swap* (identity switch) that can persist.
- **PDAF (per-track soft):** because it does not enforce one-to-one exclusivity between tracks, both tracks can partially “share” the same strong measurement, leading to *coalescence* (tracks drift toward each other).

(c) With higher clutter or lower P_D , ambiguity increases:

- **GNN** typically degrades early via wrong hard assignments and swaps (no explicit missed-detection modeling or soft uncertainty sharing).

- **PDAF** remains more stable than GNN but is vulnerable to coalescence because tracks update independently.
- **JPDA** is usually most robust among these three because it enforces mutual exclusivity: if Track 1 “claims” a measurement with high probability, Track 2’s claim is suppressed.

In short: as ambiguity grows, expect GNN to fail first (swaps), then PDAF (coalescence), while JPDA should retain better identity consistency (at higher computational cost).